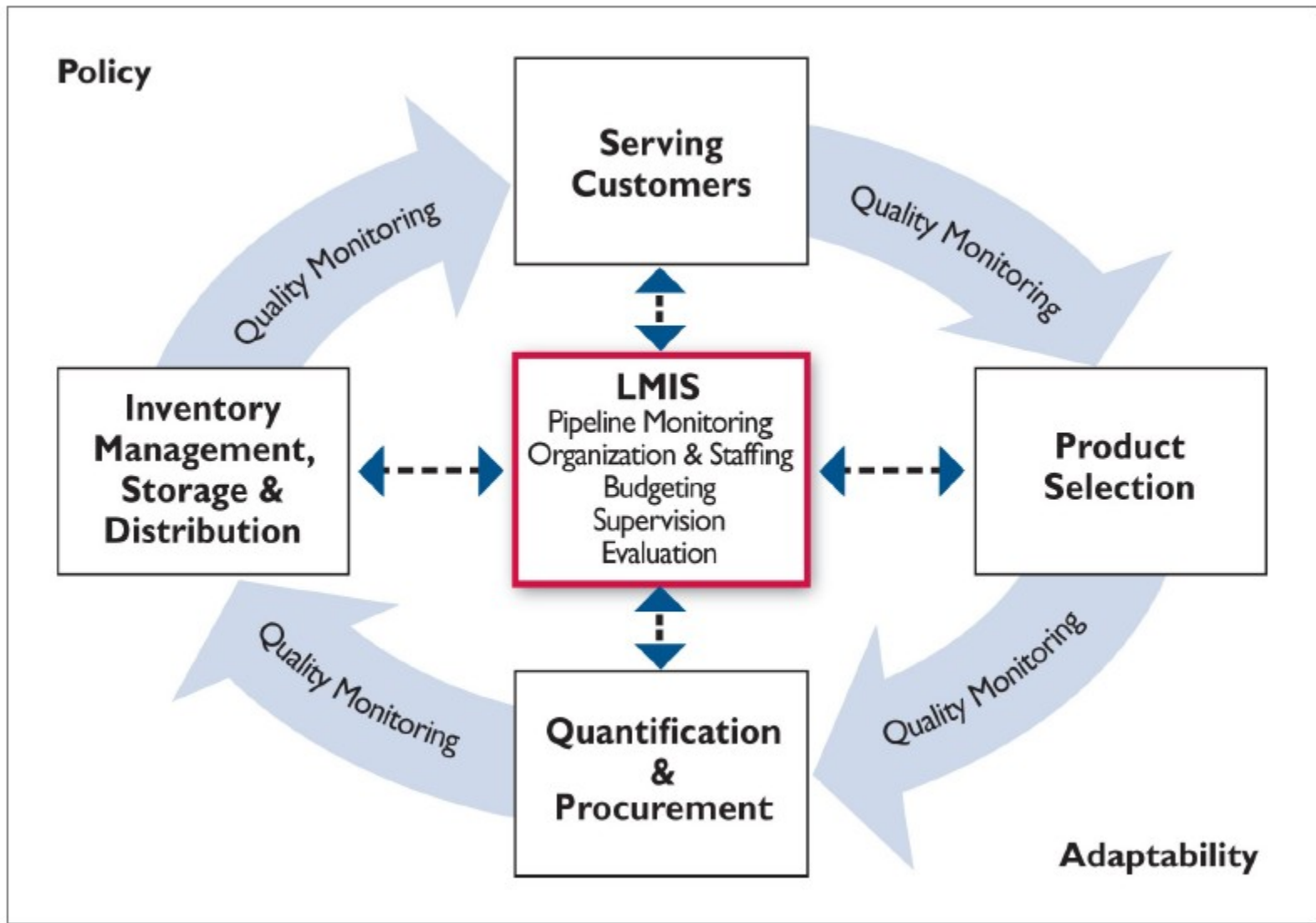


PHARMACEUTICAL SUPPLY CHAIN MANAGEMENT (PHAR 4212)

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Enabling Objectives:

At the end of this chapter, participants will be able to:

- ✓ Define rational use of medicines
- ✓ Describe the pillars of medicine use
- ✓ Discuss causes and consequences of irrational use of medicines
- ✓ Identify interventions to promote RMU

Medicine USE

- Medicine use involves a process of prescribing, dispensing and patient use.
 - Rational Medicine use
 - Irrational Medicine use

Rational Medicine use

- Rational Medicines Use (RMU) is the process of safe, effective, and economic use of medicines for diagnosis, prevention, and treatment of diseases for the benefit of the patient.

Rational Medicine use (WHO)

- **Rational Medicine use** requires that patients receive medications appropriate to their clinical needs, in doses that meet their individual requirements, for an adequate period of time, and at the lowest possible cost to them and their community.

Criteria for rational Medicine use

- **Correct Medicine;**
- *Appropriate Medicine, considering efficacy, safety, suitability for the patient, and cost;*
- **Appropriate indication-** that is, prescribe on sound medical considerations;
- **Appropriate dosage, administration, and duration of treatment;**

Criteria for rational Medicine use

- Appropriate patient that is, no contraindications exist, and the likelihood of adverse reactions is minimal;
- Correct dispensing, including appropriate information for patients about the medicines;
- Patient adherence to treatment.

Medicines Use Process

Who are the major actors in the
medicine use process?

Medicines Use Process

- Accordingly, there are three main actors in the medicines use process:
- **the prescriber, dispenser, and client.**

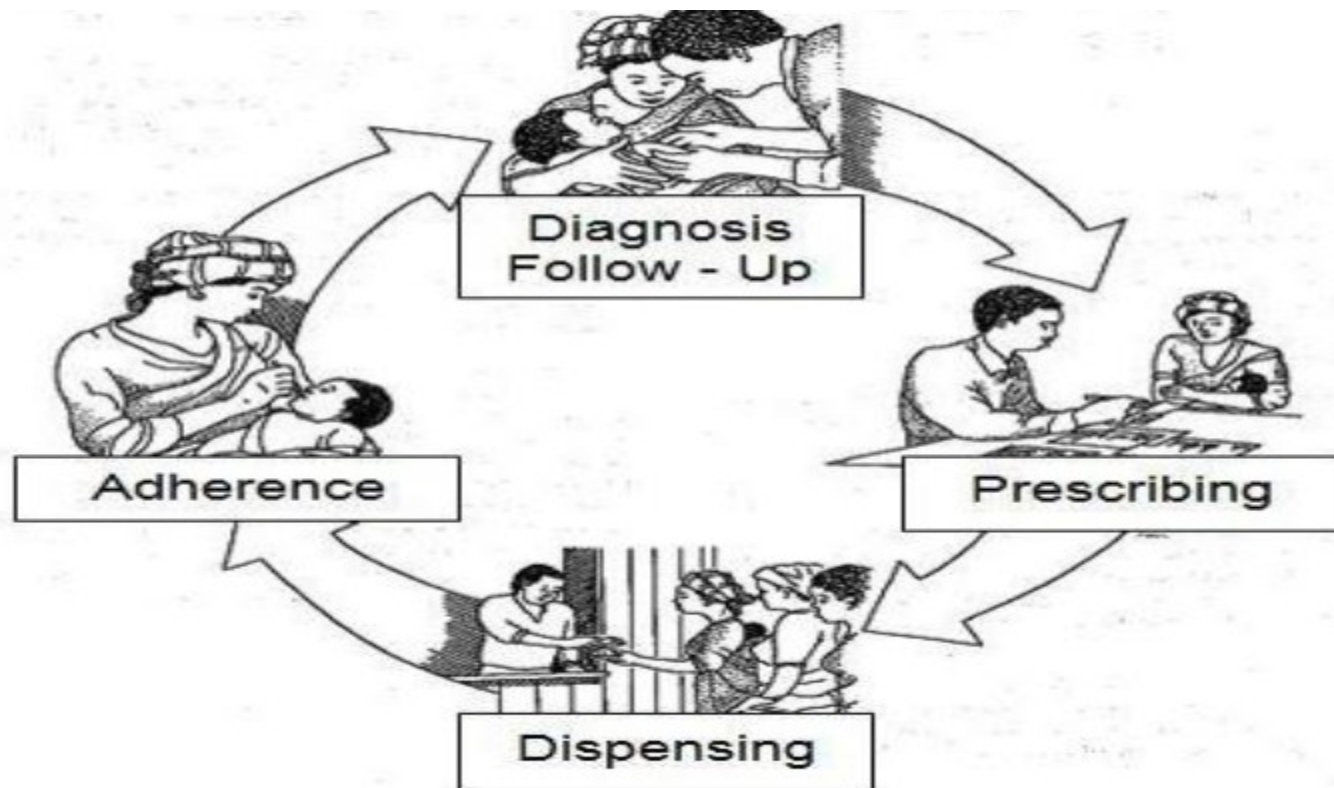


Figure 2: The medicine use process

Medicines Use Process

- The **prescriber** is responsible for prescribing a course of treatment after appropriate diagnosis of patient complaints.
- The **dispenser** is responsible for preparing and providing the medications to the patient with adequate counselling.
- The **client** is responsible for taking the medicines as per the adherence counselling provided by the prescriber and dispenser.

2. Irrational Medicine use

Irrational Medicine use

- It is inappropriate use of Medicines arising from problems in prescribing, dispensing or due to patient in adherence
- *Irrational use of Medicines may occur either due to irrational Prescribing, irrational dispensing, and irrational patient use.*

Common forms and causes of irrational use of medicines

Common forms and causes of irrational prescribing

Extravagancy:

- An expensive brand drug is used where less expensive equivalents are available

Over prescribing:

- The drug is not needed
- The dose is too large
- The treatment period is too long
- The quantity prescribed is too much for the current course of treatment

Common forms and causes of irrational prescribing

Incorrect prescribing:

- When the drug is given for an incorrect diagnosis
- Wrong drug is selected for the indication;
- Prescription is written inappropriately
(not legible, information not complete)

Common forms and causes of irrational prescribing

Multiple Prescribing:

- Using two or more medications when one or two would achieve the same effect;

Under Prescribing:

- When needed medications are not prescribed or the dosage is inadequate;
- Length of treatment is too short

Common Forms and Cause of Irrational Dispensing Practices

- Misreading the prescription
- Prescription evaluation error
- Errors during verbal communication
- Picking error, i.e., picking the next medicine
- Counting error, i.e., dispensing the wrong quantity of medicines
- Expiry error, i.e., dispensing expired medicines

Common forms of Irrational Patient Use and their Causes

- Medicines are not used according to the information provided by the prescriber and dispenser
- Self-medication with prescription medicines and wrong use of antibiotics
- Using two or more medicines simultaneously and unnecessarily
- Unsafe traditional medicines
- Overuse of injections,
- Use of needlessly expensive medicines

What do you think, what are the factors that cause irrational Medicine use?

Irrational Medicine use arising from

✓ **Prescribers and dispensers problem**

 Lack of training

 Shortage of materials

 Inadequate information

 Absence of supervision

 Financial interest

Irrational Medicine use arising from

✓ **Patient and community belief**

 Cultural and religion belief

 Awareness

 Short consultation time

Irrational Medicine use arising from

✓ **health system problem**

 Poor Medicine storage

 Expiry Medicines

 Limited Diagnostic involved in Medicine prescribing

 Absence of standard treatment guidelines for the common health problems

Consequences of irrational medicines use

Irrational use of medicines may lead to ineffective & unsafe treatment

- ✓ Exacerbation or prolongation of illness & harm to patient,
- ✓ Increase the cost of treatment,
- ✓ Antimicrobial resistance and
- ✓ Adverse drug events

Interventions to Promote Rational Medicines Use

Strategies to improve Medicine use

1. Educational strategies: TO PERSUADE
2. Managerial strategies: TO GUIDE
3. Economic strategies
4. Regulatory strategies: TO ENFORCE

Educational strategies: TO PERSUADE

- **Training of Prescribers**

- Continuing education (in-service)
- Educational Supervisory visits
- Group lectures, seminars and workshops

- **Printed Materials**

- Clinical literature and newsletters
- Treatment guidelines and Medicine formularies
- Illustrated materials (flyers, leaflets)

Managerial Strategies: TO GUIDE

- **Monitoring, supervision and feedback**
 - Hospital Medicine and therapeutic committees
 - District health teams
 - Government inspectorate
 - Professional organizations
 - Self-assessment
- **Selection, Procurement and Distribution**
 - Limited procurement lists
 - Medicine utilization review and feedback
 - Medicine committees

Economic Strategies

- **Price setting**
- **Insurance**

Regulatory Strategies: TO ENFORCE

- **Medicine selection**

- Banning unsafe or ineffective Medicines

- **Prescribing and Dispensing Controls**

- Level-of-use prescribing restrictions (health post, health center, hospital)
- Restrictions on who can prescribe or dispense
- Requirements for generic prescribing